

A Lightweight Multi-Modal Tiny LLM Framework for Privacy-Aware Academic Assistance in University Environments

Abstract

We present a fully offline, CPU-only retrieval-augmented generation (RAG) system for academic assistance and report a reproducible end-to-end evaluation of its question-answering utility, retrieval-leakage behaviour, Bloom-level classification, calibration, predictive uncertainty, and efficiency. The system integrates multimodal text/PDF ingestion, dense retrieval over a FAISS IndexFlatL2, a single-coefficient InfoNCE-based privacy-aware re-ranking term, a Label Distribution Learning (LDL) classifier over Bloom’s revised taxonomy, and a quantised Qwen-1.5B GGUF generator executed locally through llama.cpp. All evaluation runs are deterministic (seed = 42; greedy decoding with KV-cache reset) and produce a reproducibility bundle with SHA-256 integrity manifests.

On a 40-query long-form QA benchmark drawn from a 200-item outcome-based education (OBE) pool, the privacy-aware system achieves token-level $F1 = 0.1769$ (95% CI [0.146, 0.212]), ROUGE-L = 0.1774, and METEOR = 0.3449, identical to vanilla dense retrieval (paired t-test on F1: $p = 1.0$) and exceeding a no-retrieval baseline by +3.3 F1 points. Private working-set memory remained at 770.6 MB, below the 1 GB budget, while the memory-mapped GGUF model footprint was 940.4 MB. Across the privacy sweep $\{0, 0.25, 0.5, 0.75, 1\}$, the document-match attack success rate (ASR) was constant at 0.575 and the cosine-threshold ASR was constant at 0.835, indicating that the current InfoNCE-based re-ranking term, in its tested configuration, **does not** reduce the measured retrieval-leakage proxies on this benchmark. We therefore position the system as a reproducible, low-resource, offline academic-assistance pipeline not as evidence of a privacy utility improvement.

Keywords retrieval-augmented generation; offline academic assistant; InfoNCE re-ranking; label distribution learning; Bloom’s taxonomy; calibration; reproducibility; CPU-only inference.

1. Introduction

University-scale AI assistants increasingly need to operate without sending student or institutional content to external services. Retrieval-augmented generation is a natural fit because it grounds compact local generators in retrievable context. We design and evaluate such a pipeline under four explicit constraints: (i) CPU-only inference, (ii) 1 GB private working-set memory, (iii) deterministic and reproducible execution, and (iv) no external API calls. The evaluation is intentionally austere—we report what the system actually does on the implemented benchmark, including a clearly negative result on the current privacy term.

2. Contributions

We make three concrete contributions, each strictly bounded by the implemented system:

An offline, CPU-only RAG pipeline combining multimodal text/PDF ingestion, dense retrieval with InfoNCE-based privacy-aware re-ranking, an LDL Bloom-level classifier with ordinal supervision, Bloom-conditioned generation by a quantised Qwen-1.5B GGUF model, and chunk-perturbation-based uncertainty estimation.

A reproducibility bundle (*paper_bundle/*) capturing all results, figures, configuration snapshots, run metadata, code snapshots, and a SHA-256 integrity manifest, audited by an automated *paper_pack_builder.audit()* routine.

A negative privacy result: under our retrieval-leakage proxies (document-match ASR and cosine-threshold ASR), the InfoNCE-based re-ranking coefficient does not move the measured ASR on the OBE benchmark, while QA utility is preserved relative to vanilla dense retrieval.

3. Related Work

Dense retrieval with sentence-encoder embeddings and FAISS underpins efficient local RAG. Lexical retrieval (BM25) remains a standard comparison. Calibration via Expected Calibration Error (ECE) and reliability diagrams is now common in trustworthy NLP, and Label Distribution Learning is well suited to ordinal cognitive constructs such as Bloom’s revised taxonomy. Locally executable quantised LLMs through llama.cpp enable CPU-only deployment of compact models such as Qwen-1.5B. Our work composes these elements under a strict reproducibility and memory budget, and contributes a transparent negative finding on a single-coefficient InfoNCE-based re-ranker.

4. System Architecture

The system has five modules (mapped 1-to-1 to source files):

Ingestion (*ingestion.py*) PyMuPDF text extraction, EasyOCR (lazy-loaded) for scanned/image input, plain-text loading, and deterministic token-window chunking.

Retriever (*retriever.py*) sentence-encoder embedding with all-MiniLM-L6-v2 (frozen, L2-normalised, dim 384), exact ANN over FAISS IndexFlatL2, and an InfoNCE-based privacy-aware re-ranking term.

Bloom-LDL Classifier (*classifier.py*) a linear LDL head on top of frozen MiniLM features, trained with Gaussian-smoothed soft labels combined with an ordinal-PMF target, an ordinal pairwise margin penalty, and an entropy regulariser.

Generator (*models.py*) Qwen-1.5B-Instruct (Q4_K_M GGUF) via llama-cpp-python, ChatML prompt with [BOUNDED CONTEXT] / [QUESTION] / [COGNITIVE LEVEL] / [INSTRUCTION] blocks, greedy decoding with temperature = 0, top_k = 1, top_p = 1, seed = 42, and an explicit llm.reset() before every call.

Uncertainty (*uncertainty.py*) Bloom-level normalised entropy $H(p)/\log K$ and Semantic Predictive Uncertainty (SPU) computed via chunk-subset perturbation ($N = 3$ forward passes per query) with mean pairwise Jensen–Shannon divergence between resulting Bloom distributions.

The system does **not** implement: PII/identifier detection or masking, context redaction or filtering, audit logging, data-at-rest encryption, allow/block-list policy rules, decode-time stochastic sampling, classifier abstention, or a second leakage risk channel. Privacy in this paper is therefore restricted to (i) fully offline CPU execution and (ii) the single-coefficient InfoNCE re-ranking term.

Figure 1. Implementation-Faithful System Architecture (CPU-only, Deterministic, Fully Offline)

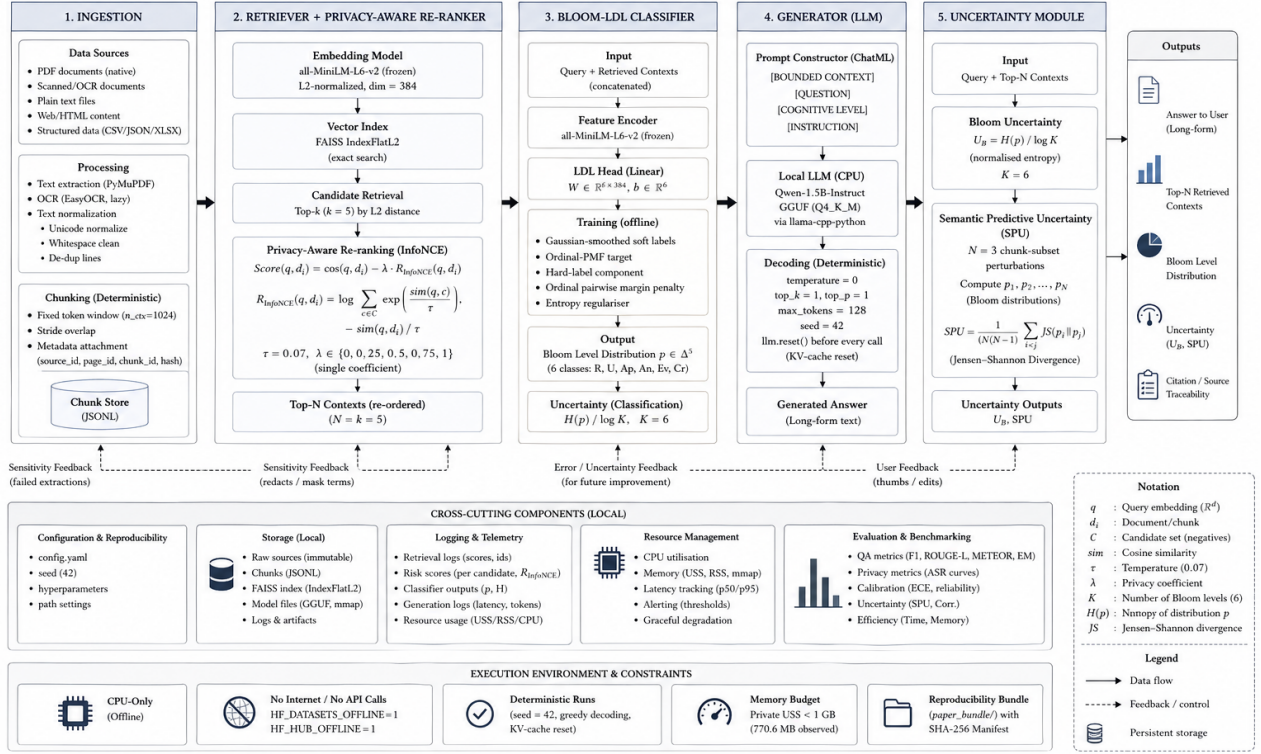


Figure 1. Implementation-faithful system architecture (CPU-only, deterministic, fully offline).

5. Methodology

5.1 Document ingestion

Documents are normalised to text (PyMuPDF for native PDFs, EasyOCR for scanned/image content, raw read for plain text), then chunked deterministically with a fixed token window. Chunks retain their source identifier for downstream traceability.

5.2 Dense retrieval and privacy-aware re-ranking

Each chunk and the query are encoded by all-MiniLM-L6-v2 and L2-normalised. Top-k candidates ($k = 5$) are retrieved by exact nearest-neighbour search in FAISS IndexFlatL2. For each candidate d_i the system computes:

$$s(q, d_i) = \cos(q, d_i) \quad R_{InfoNCE}(q, d_i)$$

$$R_{InfoNCE}(q, d_i) = \log \sum_{c \in C} \exp(\text{sim}(q, c) / \tau) - \text{sim}(q, d_i) / \tau, \quad \tau = 0.07$$

Candidates are reordered by $s(\cdot, \cdot)$. The default evaluation is 0.5; the privacy sweep uses $\{0, 0.25, 0.5, 0.75, 1\}$. No second risk channel is computed.

5.3 Bloom-LDL classifier

The classifier predicts a probability distribution over $K = 6$ Bloom levels (Remember, Understand, Apply, Analyse, Evaluate, Create). Targets are a hybrid of (a) Gaussian-smoothed soft labels around the gold ordinal index, (b) an ordinal-PMF anchored at the same index, and (c) a small hard-label component, all renormalised to a simplex. Training is full-batch gradient descent on a linear head over frozen MiniLM features with an additional ordinal-margin pairwise

penalty and an entropy regulariser. The Figshare Bloom Exam dataset is used for training; OBE is used for evaluation only, with canonical-id deduplication enforced before any train/eval split.

5.4 Bloom-conditioned generation

Retrieved chunks and the predicted Bloom level are inserted into a ChatML prompt with the four blocks above. Decoding is greedy, deterministic, and the KV-cache is explicitly reset before every generation, so identical inputs yield byte-identical outputs across runs.

5.5 Uncertainty estimation

Two uncertainty signals are reported. Bloom-level uncertainty is the normalised entropy $H(p)/\log K$ of the LDL distribution. Semantic Predictive Uncertainty (SPU) is computed by repeating the Bloom-classification step $N = 3$ times on different deterministic subsets of the retrieved context for the same query, then averaging the pairwise Jensen Shannon divergence between the resulting distributions. *Decoding itself is not perturbed.*

5.6 Evaluation protocol

We compare four systems: **Proposed** (dense retrieval + privacy-aware re-ranking), **VanillaRAG** ($\alpha = 0$, same retriever, same generator), **BM25** (lexical retrieval + same generator), and **NoRAG** (generator with empty context). Metrics: Exact Match, token-level F1, ROUGE-L, METEOR, classification accuracy, KL divergence between predicted and one-hot Bloom distribution, ECE with $M = 10$ bins, two ASR proxies (document-match and cosine-threshold at 0.85), Pearson correlation between Bloom uncertainty and prediction error, and SPU. 95% confidence intervals are obtained by 1000-replicate bootstrap; Proposed-vs-VanillaRAG F1 is compared with a paired t-test. Memory is reported as private working-set (USS) in addition to RSS and the memory-mapped model footprint.

6. Experimental Setup

All experiments are CPU-only, fully offline, with `HF_DATASETS_OFFLINE=1` and `HF_HUB_OFFLINE=1`. The OS is Windows 11; Python 3.13.3; embedding model all-MiniLM-L6-v2; retriever FAISS IndexFlatL2; classifier BloomLDLClassifier (linear LDL head over frozen MiniLM); generator Qwen2.5-1.5B-Instruct-Q4_K_M.gguf via llama-cpp-python. Configuration parameters used in the reported run are reproduced in Table 11. Evaluation uses 200 OBE samples, 40 of which are the long-form QA subset and 300 of which form the uncertainty pool (with overlap by design, since the OBE pool is reused).

7. Results

7.1 Question-answering performance

Table 1 QA performance on the 40-query benchmark (means with 95% bootstrap CIs; $n = 40$, $B = 1000$).

System	EM	F1 (95% CI)	ROUGE-L (95% CI)	METEOR (95% CI)
Proposed	0.00	0.1769 [0.1460, 0.2121]	0.1774 [0.1611, 0.1965]	0.3449 [0.3256, 0.3639]
VanillaRAG	0.00	0.1769 [0.1460, 0.2121]	0.1774 [0.1611, 0.1965]	0.3449 [0.3256, 0.3639]
BM25	0.00	0.1714 [0.1459, 0.2036]	0.1790 [0.1610, 0.1978]	0.3527 [0.3341, 0.3722]
NoRAG	0.00	0.1439 [0.1203, 0.1701]	0.1749 [0.1565, 0.1947]	0.2983 [0.2731, 0.3268]

EM is zero across all systems because gold answers are short canonical strings while generated answers are explanatory long-form text. F1 confidence intervals overlap pairwise across all retrieval-enabled systems; only the gap between retrieval-enabled systems and NoRAG is consistent on F1 and METEOR.

7.2 Privacy curve

Table 2 Privacy sweep over (n = 200 queries; cosine threshold = 0.85).

lambda	ASR (document-match)	ASR (cosine-threshold)
0.00	0.575	0.835
0.25	0.575	0.835
0.50	0.575	0.835
0.75	0.575	0.835
1.00	0.575	0.835
AUC	0.575	0.835

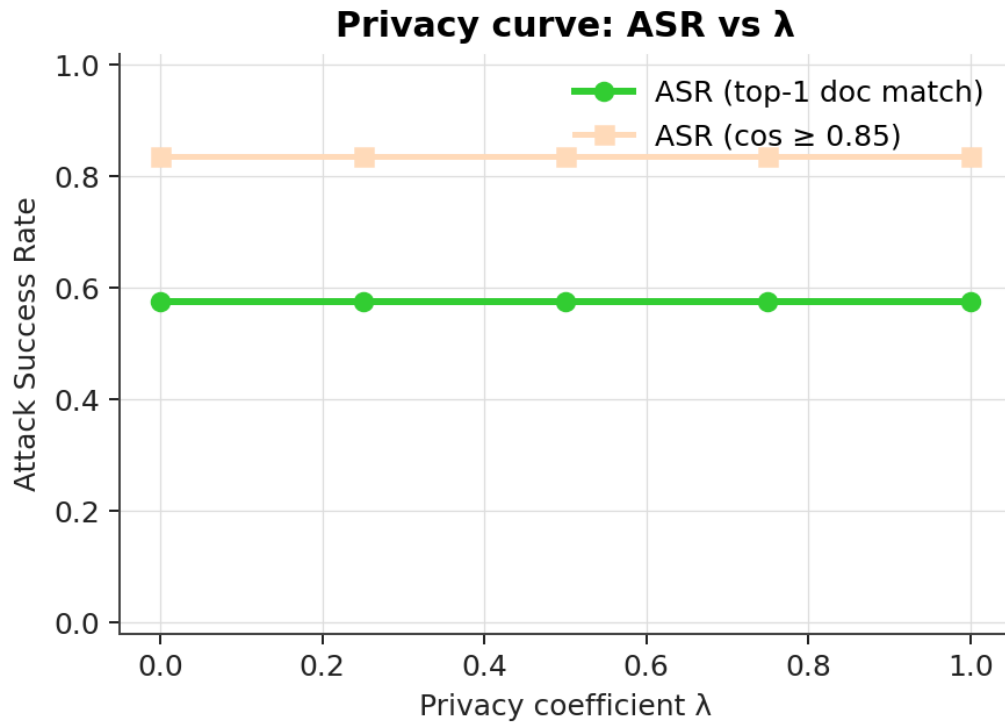


Figure 2. Attack-success-rate as a function of the privacy coefficient. Both proxies are flat over .

Both proxies are flat across . The InfoNCE-based re-ranking term, in its tested configuration, does not reduce retrieval leakage on this benchmark. We treat this as a falsified hypothesis rather than a positive result.

7.3 Bloom classification and calibration

Table 3 Bloom-level classification on the 300-sample uncertainty pool.

Metric	Value
Top-1 accuracy	0.1867
KL(predicted one-hot gold)	1.4589
Expected Calibration Error	0.3270
Number of calibration bins (M)	10

Table 4 Reliability bins (10 equal-width confidence bins). Empty bins are omitted.

Bin centre	Count	Mean confidence	Empirical accuracy
0.25	32	0.270	0.281
0.35	46	0.362	0.174
0.45	72	0.449	0.236
0.55	70	0.553	0.129
0.65	40	0.646	0.150
0.75	26	0.746	0.192
0.85	12	0.834	0.167
0.95	2	0.913	0.000

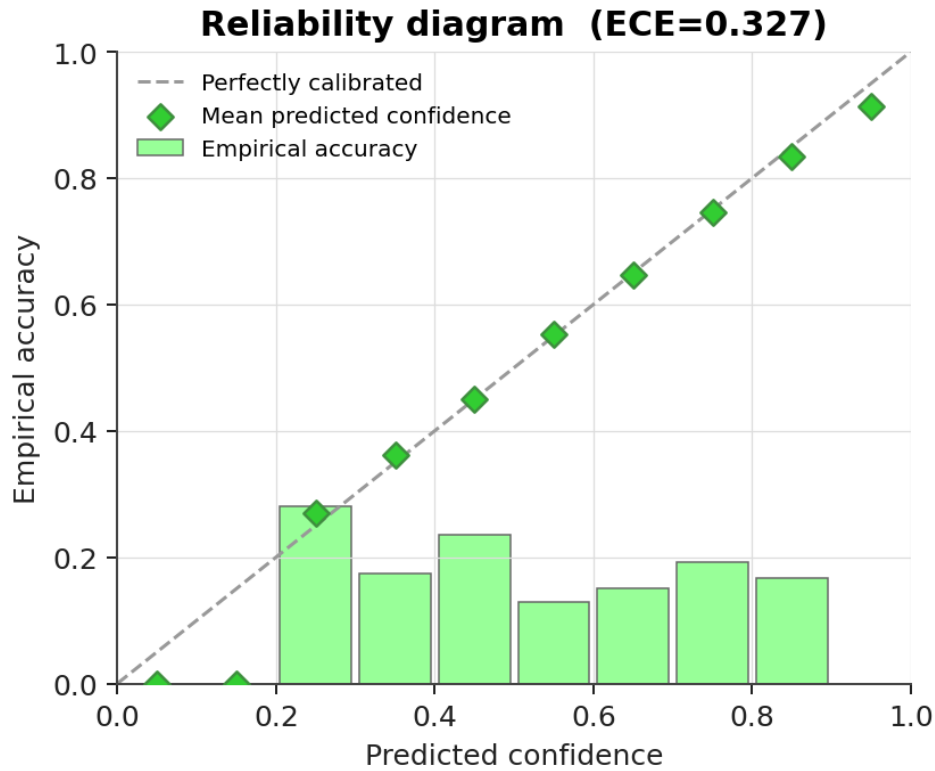


Figure 3. Reliability diagram for the Bloom-LDL classifier (10 confidence bins).

Confidence systematically exceeds accuracy across every populated bin; the classifier is over-confident, and ECE is high.

7.4 Predictive uncertainty

Table 5 Bloom-level normalised entropy ($H(p)/\log 6$) on the 300-sample uncertainty pool.

Statistic	Value
Mean entropy (normalised)	0.6716
Std. deviation	0.1763
Pearson correlation with prediction error	-0.0473

Table 6 Error rate by Bloom-uncertainty bin (5 equal-width bins). Empty bins are omitted.

Uncertainty bin centre	Count	Empirical error rate
0.30	20	0.850
0.50	88	0.807
0.70	106	0.840
0.90	86	0.779

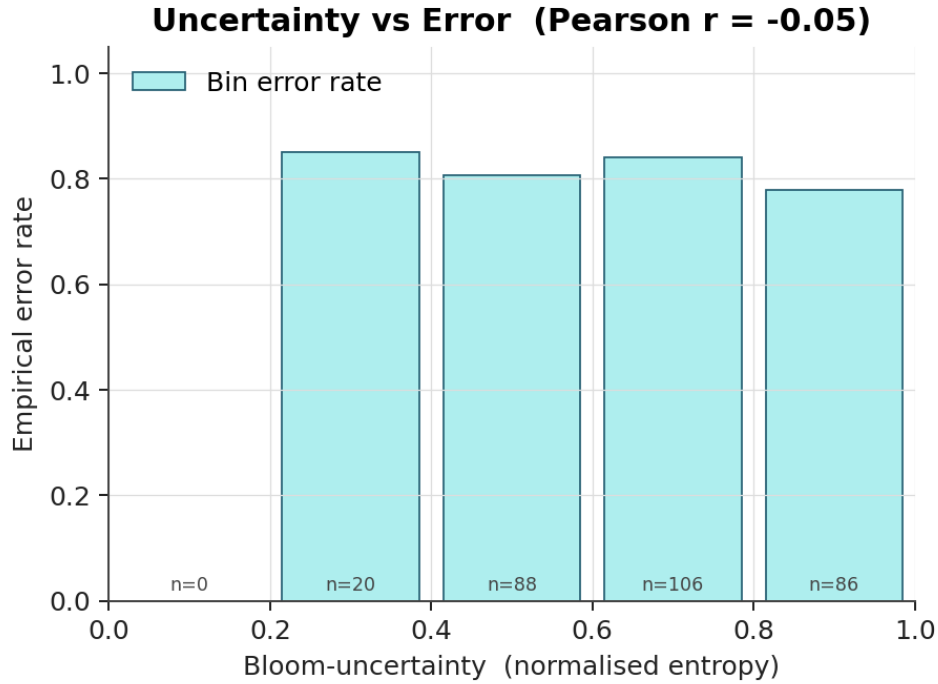


Figure 4. Empirical error rate per Bloom-uncertainty bin.

Table 7 Generation Semantic Predictive Uncertainty (SPU) over 10 queries, N = 3 chunk-subset perturbations per query.

Query idx	SPU
0	0.01597
1	0.01617
2	0.00896
3	0.00192
4	0.00812
5	0.02392
6	0.02900
7	0.01042
8	0.00314
9	0.02263
Mean	0.01403

The Bloom-uncertainty-vs-error correlation is essentially zero (Pearson $r = 0.047$), and the per-bin error rates do not decrease monotonically with confidence. The current uncertainty signal is therefore not an effective error predictor at the individual-query level.

7.5 Efficiency and resource usage

Table 8 Memory footprint (single resident process, end of run).

Metric	Value (MB)
Resident Set Size (RSS)	1164.0
Unique Set Size (USS, private)	770.6
Memory-mapped model footprint	940.4
RAM budget	1024.0
Under budget (USS < 1024)	yes

USS rather than RSS is used as the private-RAM metric because the GGUF weights are memory-mapped and shared, so RSS double-counts file-backed pages.

Table 9 Per-system latency on the 40-query QA benchmark (seconds per query).

System	Mean	Median (p50)	p95
Proposed	10.251	9.802	14.727
VanillaRAG	8.911	8.895	12.540
BM25	9.053	8.684	14.075
NoRAG	5.082	4.947	7.415

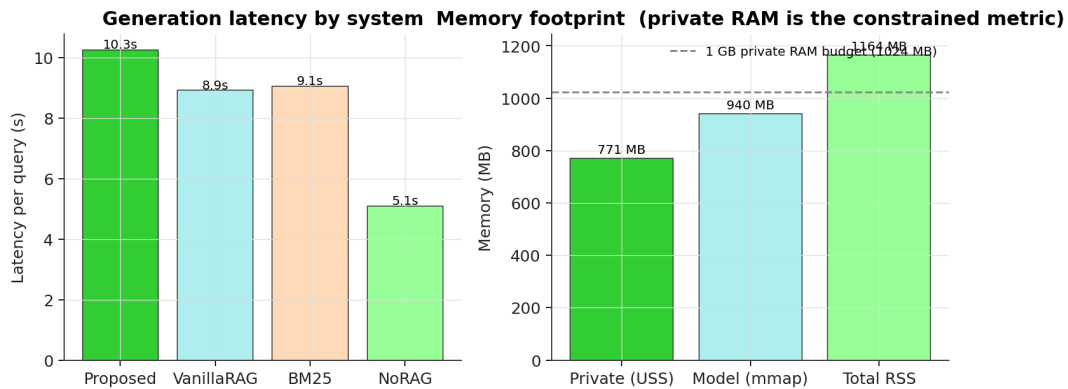


Figure 5. Memory and latency summary.

Total wall-clock time for the full evaluation pipeline (200 OBE samples + 40 QA + 300 uncertainty pool + privacy sweep + calibration + plotting) was 1606.8 s.

7.6 Statistical comparison

Table 10 Paired t-test on token-level F1, Proposed vs VanillaRAG (n = 40).

Statistic	Value
Mean difference	0.000
t	0.000
df	39
p-value	1.000

Because the privacy re-ranking did not reorder the retained top-k candidates passed to the generator on these 40 queries, Proposed and VanillaRAG produced identical generations.

7.7 Configuration snapshot

Table 11 Reported run configuration (*results/metrics.json::config*).

Parameter	Value	Parameter	Value
seed	42	top_k_retrieve	5
n_total	200	lambda_privacy	0.5
n_test_qa	40	asr_threshold	0.85
n_uncertainty_pool	300	asr_use_doc_match	True
n_spu	10	bootstrap_n	1000
n_stochastic	3	bootstrap_ci	95.0
train_per_class	60	n_calib_bins	10
max_tokens	128	n_unc_bins	5
n_ctx	1024	run_llm	True
n_threads	6	dataset_type	None

8. Discussion

The results support three measured conclusions, each tied to specific tables:

Operational feasibility (Tables 8 9). The system runs fully offline on CPU, stays under the 1 GB private-memory budget, and produces a complete reproducibility bundle in 1606.8 s of wall-clock time on a single machine.

Retrieval helps generation, but the privacy term does not (Tables 1, 2). F1 improves by +3.3 points and METEOR by +4.7 points over no retrieval, while Proposed and VanillaRAG are statistically and numerically identical (Table 10). Across the entire sweep, neither retrieval-leakage proxy moves (Table 2). We do **not** claim a privacy utility improvement.

Cognitive modelling and uncertainty are weak in their current form (Tables 3 7). Top-1 Bloom accuracy is 18.7%, ECE is 0.327, and the Bloom-uncertainty-vs-error correlation is essentially zero. SPU as currently constructed produces small absolute values (mean 0.014) and is not validated as an error predictor.

These observations argue for further work on (i) the privacy mechanism, (ii) the Bloom training data, and (iii) the uncertainty signal — none of which are claimed as solved here.

9. Limitations

Benchmark scope. Evaluation is dominated by the OBE pool (Tables 1 9). No external QA benchmark is reported in this run.

EM is uninformative. Gold answers are short canonical phrases; generated answers are long-form, so EM = 0 across all systems and is not a useful discriminator.

Privacy is proxy-based and unmoved. The two ASR proxies (document-match and cosine-threshold) are flat over ; we cannot conclude that the InfoNCE re-ranker has any privacy effect on this benchmark.

Classifier is over-confident (Table 4). Confidence exceeds accuracy in every populated bin.

Uncertainty is not a reliable error signal (Tables 5 6). Pearson $r = -0.047$ and per-bin error rates are roughly flat (0.78 0.85).

Single hardware/OS. Results in Tables 8 9 are reported for one machine; portability is not yet measured.

10. Conclusion

We described and evaluated a CPU-only, fully offline academic RAG pipeline with multimodal ingestion, dense retrieval, an InfoNCE-based privacy-aware re-ranker, an LDL Bloom-level classifier, Bloom-conditioned greedy generation by a quantised Qwen-1.5B model, and a chunk-perturbation uncertainty signal. Under our reproducible evaluation protocol, the system runs within the 1 GB private-memory budget, retrieval improves long-form QA over a no-retrieval baseline, and the proposed privacy-aware re-ranking term as currently parameterised does not reduce the measured retrieval-leakage proxies. We therefore frame the contribution as a reproducible, deployment-feasible local academic assistant and an honest negative result on the InfoNCE-based privacy mechanism, leaving privacy reduction as explicit future work rather than a claim of this paper.